

Solid State Drives Mini Project

Comparative Study Between Direct Torque Control (DTC) and Field Oriented Control (FOC)

Submitted as part of Mini Project Report

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Abstract

This report presents a comparative study of two important high-performance control strategies used in AC motor drives, namely Field Oriented Control (FOC) and Direct Torque Control (DTC). Both methods are widely applied in industrial drives, electric vehicles, robotics, and traction systems due to their capability to provide precise torque and speed control. Field Oriented Control achieves decoupled control of flux and torque by transforming stator quantities into a rotating reference frame, thereby enabling smooth steady-state response and low ripple. Direct Torque Control, on the other hand, directly selects inverter switching states based on instantaneous torque and flux errors, offering a faster dynamic response with a simpler control structure.

The objective of this mini project is to compare DTC and FOC on the basis of operating principle, control structure, torque response, speed response, flux behavior, ripple content, switching characteristics, and implementation complexity. The study highlights the strengths and limitations of each technique and identifies suitable applications for both control methods.

1 Introduction

Electric drives play a major role in modern industrial and transportation systems. High-performance motor control is essential in applications where precise speed regulation, fast torque response, and efficient energy conversion are required. Among the advanced control techniques available for AC drives, Field Oriented Control (FOC) and Direct Torque Control (DTC) are the most commonly used.

Field Oriented Control is based on the principle of decoupling torque and flux components in a rotating reference frame, making the AC machine behave similarly to a separately excited DC motor. This allows independent control of torque-producing and flux-producing currents. Direct Torque Control, in contrast, controls torque and stator flux directly without requiring an inner current loop or pulse width modulation stage in its basic form.

This report compares these two techniques in order to understand their suitability for different operating requirements. The comparison is useful for selecting an appropriate control method in practical electric drive systems.

2 Objectives

The main objectives of this mini project are:

1. To study the basic principles of Field Oriented Control and Direct Torque Control.
2. To understand the control structure of both methods.
3. To compare their dynamic and steady-state performance.
4. To analyze torque ripple, speed response, and flux response in both techniques.
5. To identify the advantages, disadvantages, and practical applications of DTC and FOC.

3 Field Oriented Control (FOC)

3.1 Principle of Operation

Field Oriented Control is a vector control technique in which the stator currents are transformed from the three-phase stationary reference frame to the rotating dq reference frame. In this transformed frame, the direct-axis current i_d controls flux and the quadrature-axis current i_q controls torque. This decoupling enables independent control of torque and flux, similar to the control of a DC motor.

3.2 Block Diagram of Field Oriented Control (FOC)

The block diagram of Field Oriented Control illustrates the sequence of transformations and control actions used to achieve decoupled control of torque and flux. In this method, the measured three-phase currents are transformed into the rotating dq reference frame, where independent control of flux-producing and torque-producing current components is achieved. The generated control signals are then converted back and applied to the inverter through a PWM scheme.

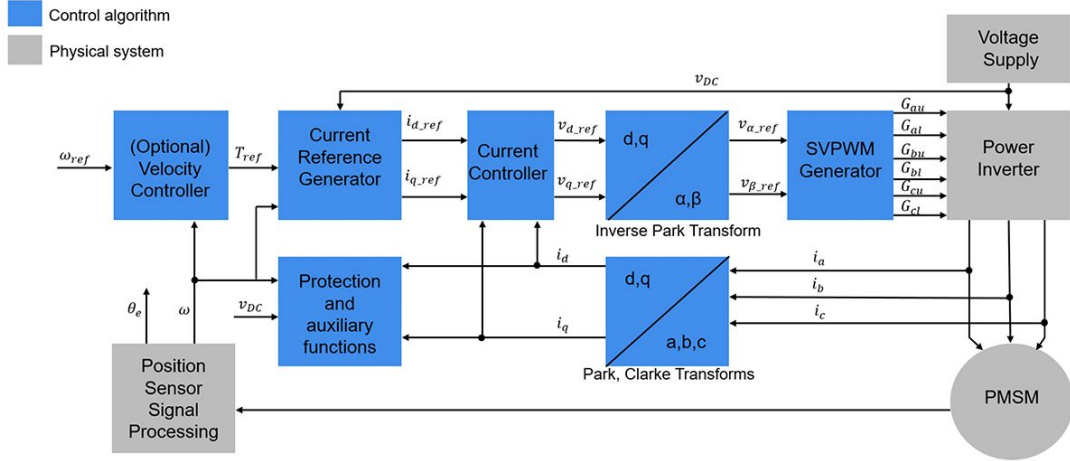


Figure 1: Block diagram of Field Oriented Control (FOC)

3.3 Control Structure

A typical FOC system consists of the following blocks:

- **Speed controller:** This block compares the reference speed with the actual motor speed and generates the reference torque-producing current component, usually denoted by i_q^* . It is generally implemented using a PI controller to reduce the speed error and ensure proper speed regulation.
- **Reference current generator:** This block generates the reference currents for the d -axis and q -axis. The d -axis current reference i_d^* is used to control the motor flux, while the q -axis current reference i_q^* is used to control torque. In many applications, i_d^* is kept constant or set to zero, depending on the motor type and operating condition.
- **Clarke transformation:** The Clarke transformation converts the measured three-phase stator currents into two-phase stationary reference frame components (α, β) . This simplifies the analysis and is the first step in transforming the variables into the rotating reference frame.

- **Park transformation:** The Park transformation converts the stationary frame components (α, β) into rotating reference frame components (d, q) using the rotor position angle. This transformation enables independent control of flux and torque components.
- **Current controllers for i_d and i_q :** These controllers compare the reference and actual d -axis and q -axis currents and generate the corresponding voltage commands v_d^* and v_q^* . They are usually PI controllers and are responsible for regulating flux and torque independently.
- **Inverse Park transformation:** This block converts the voltage commands from the rotating dq frame back into the stationary (α, β) frame so that they can be applied to the inverter.
- **PWM inverter:** The pulse width modulation block generates the switching pulses for the voltage source inverter based on the reference voltages. The inverter then supplies the required three-phase voltages to the motor.
- **Motor and feedback sensors:** The motor is the plant being controlled, and sensors provide the required feedback signals such as phase currents, rotor position, and speed. These feedback signals are essential for performing the coordinate transformations and maintaining closed-loop control.

3.4 Advantages of FOC

- Smooth torque response
- Low torque ripple
- Good steady-state performance
- High accuracy in speed control
- Suitable for precision applications

3.5 Disadvantages of FOC

- More complex implementation
- Requires coordinate transformations
- Sensitive to machine parameter variations
- Requires proper tuning of controllers

4 Direct Torque Control (DTC)

4.1 Principle of Operation

Direct Torque Control directly regulates stator flux and electromagnetic torque by selecting appropriate inverter switching states from a switching table. The control is based on the instantaneous errors between the reference and estimated values of torque and flux. In the basic DTC scheme, hysteresis comparators are used for torque and flux control, and there is no separate PWM stage.

4.2 Block Diagram of Direct Torque Control (DTC)

The block diagram of Direct Torque Control shows the direct regulation of electromagnetic torque and stator flux without the use of inner current control loops. The estimated torque and flux are compared with their reference values using hysteresis controllers, and an appropriate switching state is selected from the switching table to control the inverter. This gives DTC its fast dynamic response and relatively simple structure.

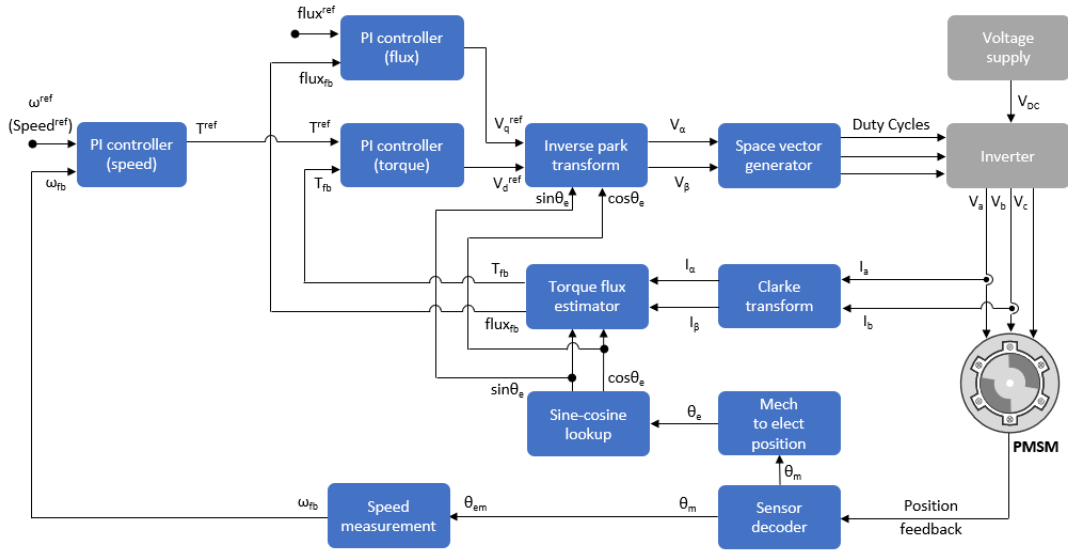


Figure 2: Block diagram of Direct Torque Control (DTC)

4.3 Control Structure

A typical DTC system consists of the following blocks:

- **Speed controller:** This block compares the reference speed with the actual motor speed and generates the reference electromagnetic torque required to reduce the speed error. It is usually implemented using a PI controller.
- **Torque and flux reference generation:** This block provides the desired reference values of torque and stator flux. The torque reference is generally obtained

from the speed controller, while the flux reference is usually maintained at a constant rated value unless flux weakening is required.

- **Torque estimator:** The torque estimator calculates the actual electromagnetic torque of the motor using measured stator currents and estimated stator flux components. This estimated torque is used for comparison with the reference torque.
- **Flux estimator:** This block estimates the magnitude and position of the stator flux vector from the measured stator voltages and currents. Accurate flux estimation is important because DTC directly controls the stator flux.
- **Hysteresis controllers:** Separate hysteresis comparators are used for torque and flux. These controllers compare the estimated values with the reference values and determine whether torque or flux needs to be increased, decreased, or maintained within a specified band.
- **Switching table:** The switching table selects the appropriate inverter switching state based on the outputs of the torque hysteresis controller, flux hysteresis controller, and the sector of the stator flux vector. It acts as the decision-making block of the DTC scheme.
- **Voltage source inverter:** The inverter applies the selected switching state to the motor terminals, thereby generating the required stator voltage vector. This directly influences the motor torque and flux.
- **Motor and feedback measurements:** The AC motor is the plant being controlled. Measurements such as stator currents, stator voltages, and rotor speed are fed back to the estimators and controller blocks to enable closed-loop control.

4.4 Advantages of DTC

- Very fast torque response
- Simple control structure
- No inner current control loop
- No modulation block required in the basic method
- Suitable for high dynamic applications

4.5 Disadvantages of DTC

- Higher torque ripple
- Variable switching frequency

- More difficult steady-state refinement
- Flux estimation sensitivity at low speed

5 Comparative Study Between DTC and FOC

Table 1 presents a comparison between Direct Torque Control and Field Oriented Control.

Table 1: Comparison between DTC and FOC

Parameter	FOC	DTC
Basic Control Philosophy	Decoupled control of torque and flux using dq transformation	Direct control of torque and stator flux using switching table
Control Complexity	Higher	Lower
Dynamic Response	Fast	Very fast
Torque Ripple	Low	Higher
Steady-State Performance	Better	Moderate
Switching Frequency	Nearly constant	Variable
Current Controllers	Required	Not required in basic DTC
PWM Requirement	Required	Not required in basic DTC
Parameter Sensitivity	Sensitive to machine parameters	Moderate, but estimation quality matters
Implementation	More complex	Comparatively simpler
Application Suitability	Precision drives, EVs, robotics	High dynamic response drives, traction, industrial drives

6 Methodology

To perform a fair comparison between DTC and FOC, both controllers should be tested under identical operating conditions using the same motor parameters, inverter configuration, and load conditions. The following procedure may be adopted:

1. Model the AC motor drive system.
2. Implement the FOC control scheme with speed and current control loops.
3. Implement the DTC control scheme with torque and flux estimation.
4. Apply the same speed reference and load torque conditions to both systems.

5. Record the transient and steady-state responses.
6. Compare the results using speed, torque, current, and flux waveforms.

7 Results and Discussion

The performance of Direct Torque Control (DTC) and Field Oriented Control (FOC) should be analyzed separately so that the characteristics of each control method can be clearly observed. The important results include speed response, electromagnetic torque response, torque ripple, stator current waveform, load disturbance response, and flux response. These plots help in evaluating both transient and steady-state performance.

7.1 Results for Field Oriented Control (FOC)

7.1.1 Speed Response of FOC

The speed response plot for FOC should show the reference speed and actual motor speed. This plot is used to study rise time, settling time, overshoot, and steady-state error under closed-loop control.

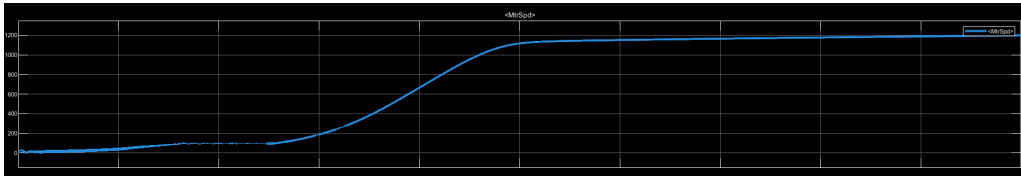


Figure 3: Speed response of Field Oriented Control (FOC)

7.1.2 Electromagnetic Torque Response of FOC

The torque response of FOC should be plotted to observe how smoothly the controller develops torque and how accurately it tracks the required operating condition.

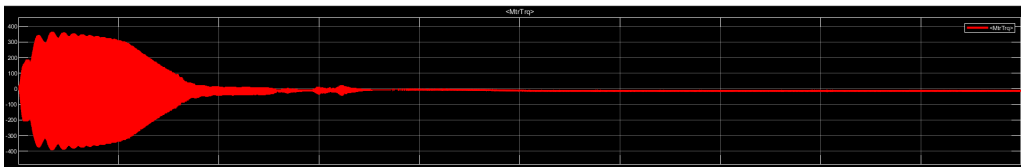


Figure 4: Electromagnetic torque response of Field Oriented Control (FOC)

7.1.3 Torque Ripple of FOC

A zoomed steady-state torque plot should be included to show the torque ripple under FOC. This is useful to demonstrate the smoother torque profile generally obtained with FOC.

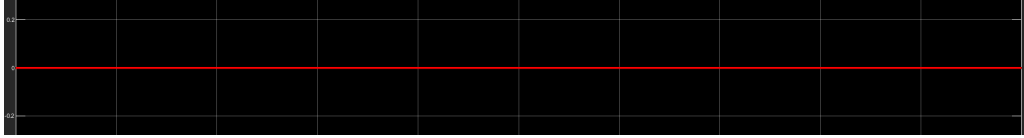


Figure 5: Steady-state torque ripple in Field Oriented Control (FOC)

7.1.4 Stator Current Waveform of FOC

The stator current waveform should be plotted to examine the quality and smoothness of the current supplied to the motor under FOC.

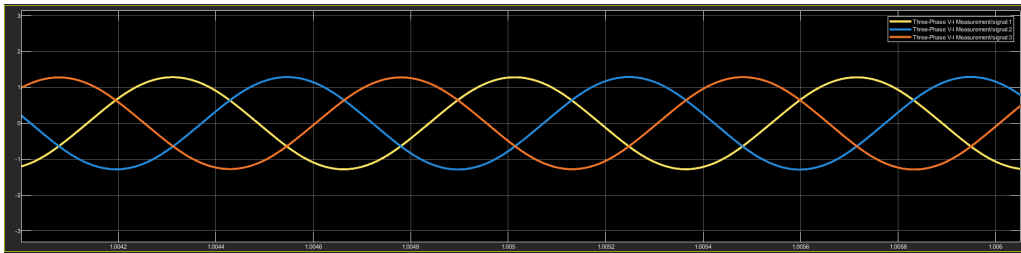


Figure 6: Stator current waveform of Field Oriented Control (FOC)

7.2 Results for Direct Torque Control (DTC)

7.2.1 Speed Response of DTC

The speed response plot for DTC should show the reference speed and actual motor speed. This plot helps in examining the speed tracking performance and transient characteristics of DTC.

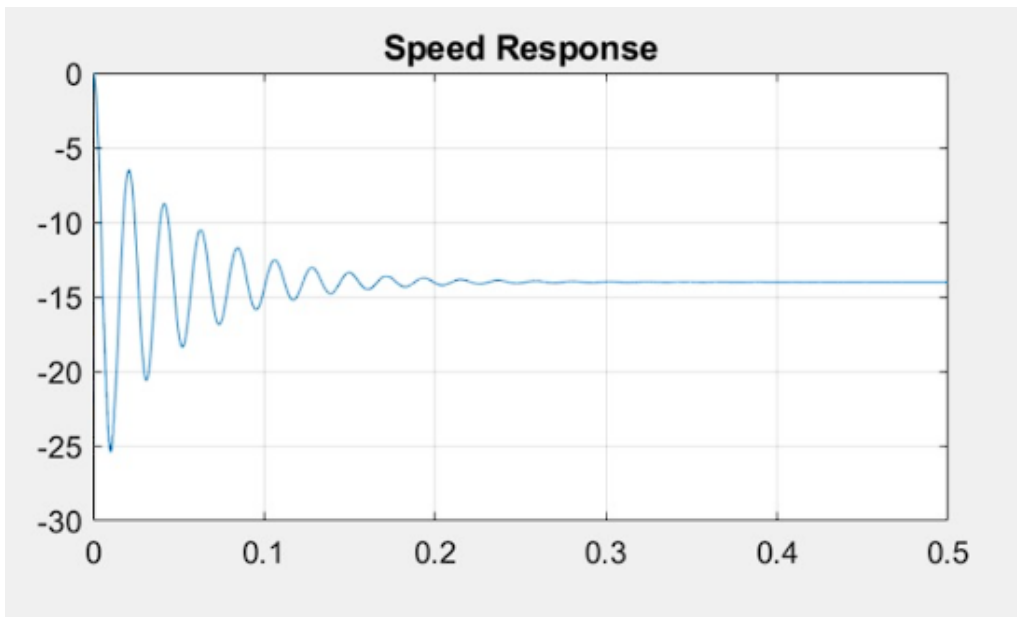


Figure 7: Speed response of Direct Torque Control (DTC)

7.2.2 Electromagnetic Torque Response of DTC

The torque response of DTC should be plotted to demonstrate its fast torque control capability. This is one of the most important plots in the comparative study.

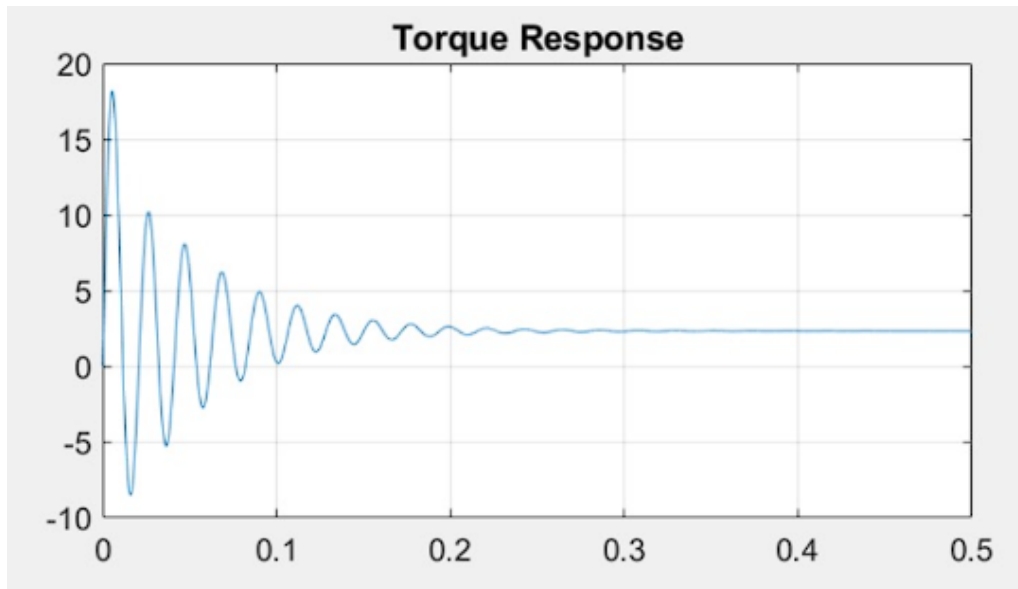


Figure 8: Electromagnetic torque response of Direct Torque Control (DTC)

7.2.3 Torque Ripple of DTC

A zoomed steady-state torque plot should be included to show the torque ripple in DTC. This helps in illustrating the relatively higher ripple commonly associated with DTC.

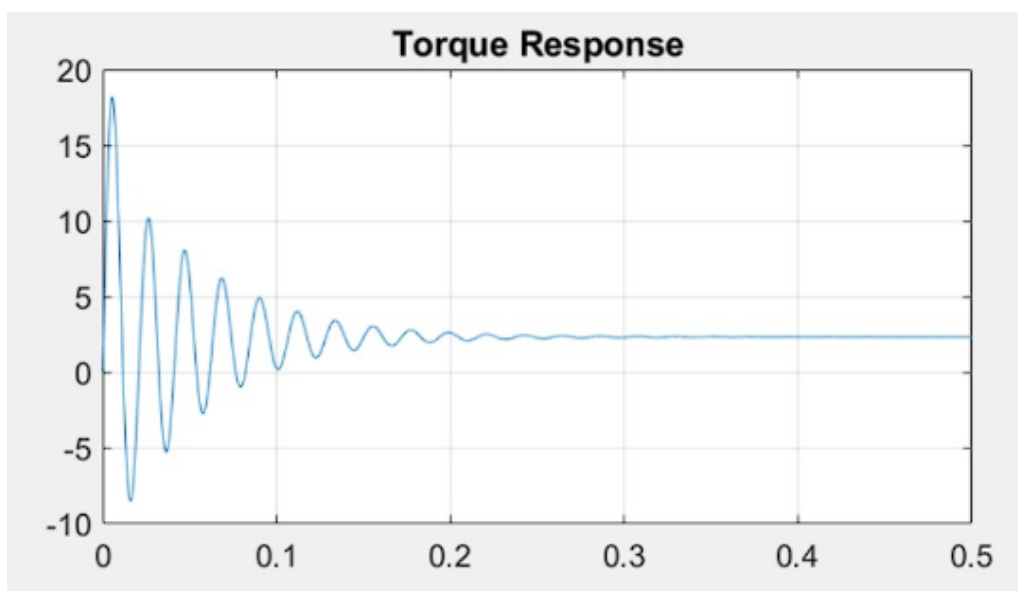


Figure 9: Steady-state torque ripple in Direct Torque Control (DTC)

7.2.4 Stator Current Waveform of DTC

The stator current waveform should be plotted to analyze the current quality under DTC operation and to observe any waveform distortion or ripple.

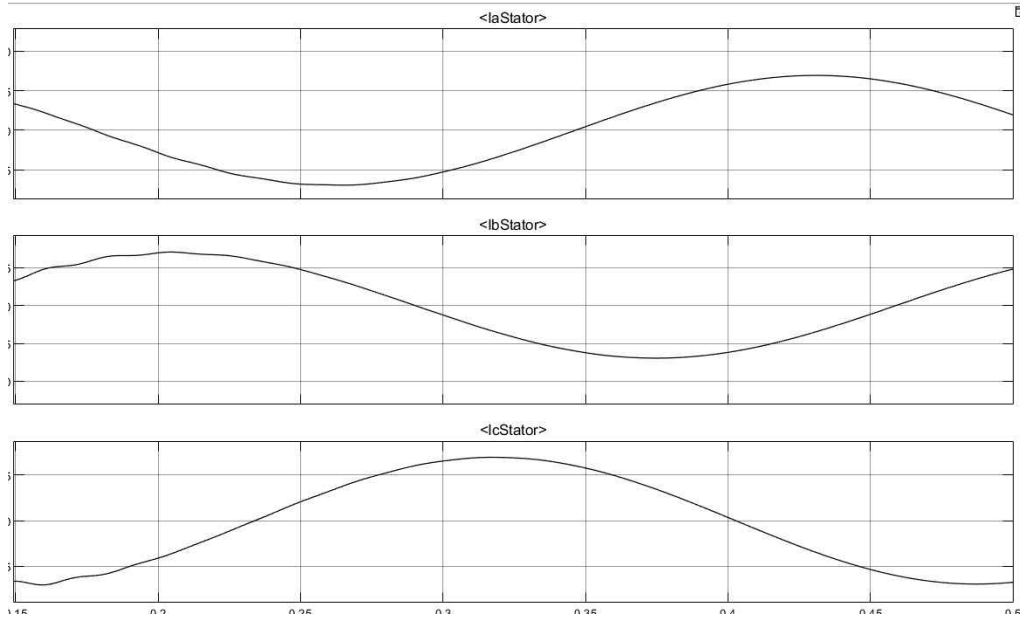


Figure 10: Stator current waveform of Direct Torque Control (DTC)

7.3 Discussion of Results

The important performance parameters to be discussed for both Direct Torque Control (DTC) and Field Oriented Control (FOC) are presented in Table 2.

Table 2: Performance parameters for discussion of DTC and FOC

Parameter	DTC	FOC
Rise time	0.000359	0.351790
Settling time	0.2877	0.15
Overshoot	682.94	25
Peak	18.2009	9.004
Peak Time (s)	0.0054	0.0023

After discussing the individual performance of DTC and FOC, the observations can be used in the comparison section to highlight the strengths and limitations of each control strategy.

8 Applications

8.1 Applications of FOC

- Electric vehicles
- Servo drives
- Robotics
- CNC machines
- Precision motion control systems

8.2 Applications of DTC

- Industrial motor drives
- Railway traction
- High-performance torque control systems
- Applications requiring rapid transient response

9 Conclusion

This report presented a comparative study between Field Oriented Control and Direct Torque Control for AC motor drives. Both techniques are capable of high-performance control, but they differ in structure, complexity, and output characteristics. FOC provides smoother torque, low ripple, and better steady-state operation, making it suitable for precision-oriented applications. DTC provides faster dynamic torque control with a simpler control structure, making it attractive for applications where quick response is required.

Thus, neither method is universally superior. The choice between DTC and FOC depends on the application requirements, available computational resources, and the acceptable trade-off between dynamic response and ripple performance.

References

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